

STAT 7650: Computational Statistics

8. Markov Chain Monte Carlo

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Markov Chain

Consider a sequence of random variables $\{X^{(t)}, t = 0, 1, \dots\}$

- **State**: possible values of $X^{(t)}$
- **State space**: set of all possible values of $X^{(t)}$

The **Markov property** assumes that the next state only depends on the present state. It is also called one-step memory.

$$P(X^{(t)} \mid X^{(0)}, \dots, X^{(t-1)}) = P(X^{(t)} \mid X^{(t-1)})$$

Hence,

$$P(X^{(0)}, \dots, X^{(n)}) = P(X^{(0)})P(X^{(1)} \mid X^{(0)}) \cdots P(X^{(n)} \mid X^{(n-1)})$$

Transition Probability

Focus on univariate, discrete-time, discrete-state-space Markov chain.

The sequence $\{X^{(t)}, t = 0, 1, \dots\}$ is a Markov chain if

$$\begin{aligned} p_{ij}^{(t)} &= P(X^{(t+1)} = j \mid X^{(0)} = x^{(0)}, \dots, X^{(t)} = i) \\ &= P(X^{(t+1)} = j \mid X^{(t)} = i) \end{aligned}$$

The **one-step transition probability** $p_{ij}^{(t)}$ is the probability that the observed state changes from state i at time t to state j at time $t + 1$.

- **Time homogeneous:** $p_{ij}^{(t)} = p_{ij}$ does not change with t
- **Transition probability matrix:** $P = (p_{ij})$ is a $s \times s$ matrix, where the state space contains s states. Each row of P must be 1.

Example: San Francisco Weather

Consider two possible conditions, wet or dry, for each day.

	observed		transition probability	
	wet today	dry today	wet today	dry today
wet yesterday	418	256	0.620	0.380
dry yesterday	256	884	0.225	0.775

The estimated transition probability matrix can be obtained from the historical data. For example,

$$P(\text{dry today} \mid \text{wet yesterday}) = \frac{256}{418 + 256} = 0.380$$

Properties of Markov Chain

A Markov chain is **ergodic** if it is irreducible, aperiodic, and all its states are nonnull and recurrent.

- A Markov chain is **irreducible** if any state j can be reached from any state i in a finite number of steps for all i and j .
- A Markov chain is **periodic** if it can visit certain portions of the state space only at certain regularly spaced intervals. State j has period d if the probability of going from state j to state j in n steps is 0 for all n not divisible by d .
- If every state in a Markov chain has period 1, then the chain is **aperiodic**.
- A state is **recurrent** if the chain returns to it with probability 1.
- A state is **nonnull** if the expected time until recurrence is finite.

Stationary Distribution

Any discrete probability distribution π such that $\pi^T P = \pi^T$ is called a **stationary** distribution for P .

- $\pi = (\pi_1, \dots, \pi_s)^T$ with $\sum_i \pi_i = 1$.
- $\pi_i = P(X^{(t)} = i)$ is the marginal probability of $X^{(t)}$
- The marginal probability of $X^{(t+1)}$ must be $\pi^T P$.
- If $X^{(t)}$ follows a stationary distribution, then the marginal distributions of $X^{(t)}$ and $X^{(t+1)}$ are identical.
- If a time-homogeneous Markov chain satisfies

$$\pi_i p_{ij} = \pi_j p_{ji}, \quad \text{for all } i, j \in \mathcal{S},$$

then π is a stationary distribution for the chain, and the chain is called **reversible**. The equation is called the **detailed balance condition**.

Ergodic Theorem

If a Markov chain with transition probability matrix P and stationary distribution π is irreducible and aperiodic, then π is unique and

$$\lim_{n \rightarrow \infty} P(X^{(t+n)} = j \mid X^{(t)} = i) = \pi_j$$

where π_j is the j th element of π . The π_j are the solution of the following set of equations,

$$\pi_j \geq 0, \quad \sum_{i \in \mathcal{S}} \pi_i = 1, \quad \pi_j = \sum_{i \in \mathcal{S}} \pi_i p_{ij}$$

If $X^{(1)}, X^{(2)}, \dots$ are realizations from an irreducible and aperiodic Markov chain with stationary distribution π , then $X^{(n)}$ converges in distribution to π , and for any function h , provided $E_\pi[|h(X)|]$ exists,

$$\frac{1}{n} \sum_{t=1}^n h(X^{(t)}) \rightarrow E_\pi[h(X)], \quad \text{almost surely as } n \rightarrow \infty.$$

Metropolis-Hastings Algorithm

The **target density** is $f(\cdot)$. Given $X^{(t)} = x^{(t)}$, the Metropolis-Hastings algorithm generates $X^{(t+1)}$ as follows.

- Sample X^* from a **proposal density** $g(\cdot | x^{(t)})$
- Compute the **Metropolis-Hastings ratio** $R(x^{(t)}, X^*)$, where

$$R(x^{(t)}, X^*) = \frac{f(X^*)g(x^{(t)} | X^*)}{f(x^{(t)})g(X^* | x^{(t)})}$$

- Sample a value for $X^{(t+1)}$ according to the following

$$X^{(t+1)} = \begin{cases} X^*, & \text{with probability } \min\{R(x^{(t)}, X^*), 1\} \\ x^{(t)}, & \text{otherwise} \end{cases}$$

- Increment t and return to step 1

Comments

- The initial value $X^{(0)} = x^{(0)}$, satisfying $f(x^{(0)}) > 0$, can be selected at random.
- When the proposal density is symmetric, the method is known as the **Metropolis** algorithm.

$$g(x^{(t)} \mid x^*) = g(x^* \mid x^{(t)})$$

- The Metropolis-Hastings algorithm constructs a Markov chain.
- Whether the chain is irreducible and aperiodic depends on the choice of proposal distribution.
- **Burn-in period**: Discard the first part of the chain, because they may depend on initial values and do not converge yet.
- It is common to start the chain from multiple starting points to check the consistent results.

Independence Chains

The proposal distribution is $g(x^* | x^{(t)}) = g(x^*)$ for a fixed density g . Each candidate value is drawn independently of the past. The Metropolis-Hastings ratio is

$$R(x^{(t)}, x^*) = \frac{f(x^*)g(x^{(t)})}{f(x^{(t)})g(x^*)}$$

Some comments.

- The resulting Markov chain is irreducible and aperiodic if $g(x) > 0$ whenever $f(x) > 0$.
- The proposal distribution g should resemble the target distribution f , and should cover f in the tail.

Independence Chain in Bayesian Inference

In Bayesian inference, let $p(\theta)$ be the prior distribution and $L(\theta | y)$ be the likelihood. We want to sample from the posterior distribution

$$p(\theta | y) \propto p(\theta)L(\theta | y),$$

Use the prior distribution as the proposal distribution. Then

$$R(\theta^{(t)}, \theta^*) = \frac{L(\theta^* | y)}{L(\theta^{(t)} | y)}$$

The Metropolis-Hastings ratio equals the likelihood ratio.

Example: Mixture of Normal Distributions

Sample $Y_1, \dots, Y_{100}$ from the mixture distribution with $\delta = 0.7$

$$\delta N(7, 0.5^2) + (1 - \delta)N(10, 0.5^2)$$

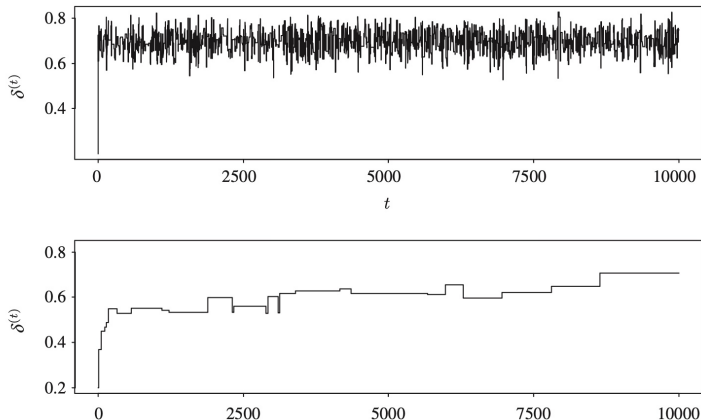
The prior distribution of δ is $\text{unif}(0, 1)$.

Compare two different independence chains.

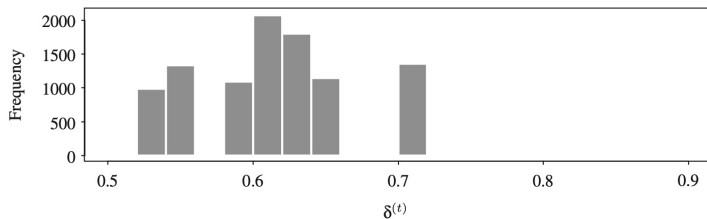
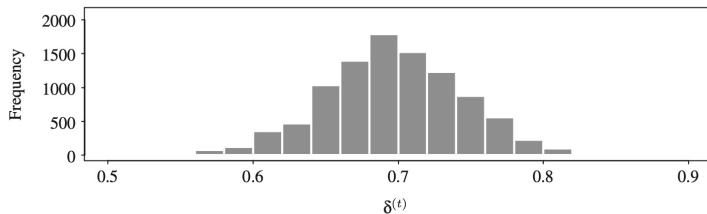
- The proposal distribution is $\text{Beta}(1, 1)$: good mixing
- The proposal distribution is $\text{Beta}(2, 10)$: poor mixing

Draw trace plot, autocorrelation plot, and histogram to assess convergence.

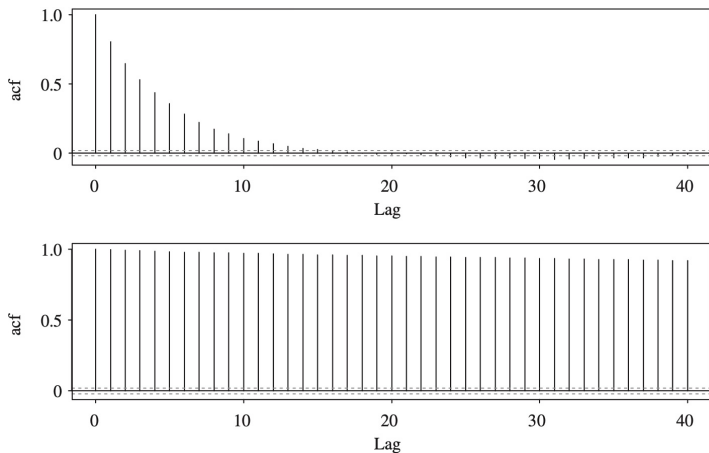
The proposal distribution is Beta(1, 1) in the top plot, and Beta(2, 10) in the bottom plot.



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Random Walk Chains

Let h be a density. For random walk chains,

- Choose the proposal distribution as

$$g(x^* | x^{(t)}) = h(x^* - x^{(t)})$$

- Or equivalently, generate a candidate value by

$$x^* = x^{(t)} + \varepsilon, \quad \text{where } \varepsilon \sim h(\varepsilon)$$

Common choices include

- uniform distribution over a ball centered at the origin
- a scaled standard normal distribution
- a scaled t -distribution

If the support region of f is connected and h is positive in a neighborhood of 0, the resulting chain is irreducible and aperiodic.

Example: Mixture Distribution

Generate a candidate value by

$$\delta^* = \delta^{(t)} + \varepsilon, \quad \text{where } \varepsilon \sim \text{unif}(-a, a).$$

There is a chance the proposed value may fall outside $(0, 1)$.
Consider reparameterization

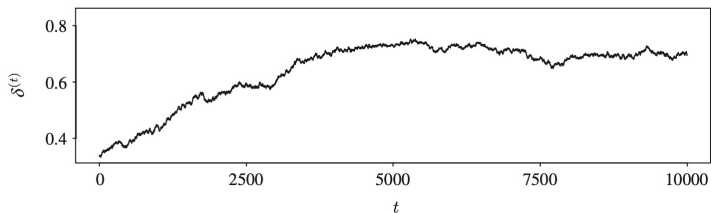
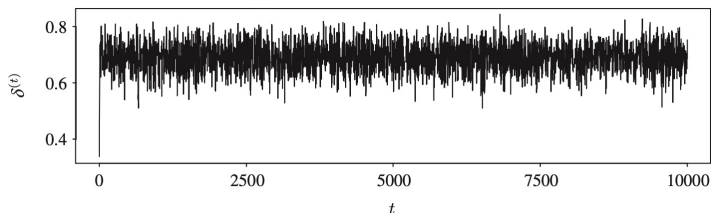
$$U = \text{logit}(\delta) = \log(\delta/(1 - \delta))$$

The Metropolis-Hastings ratio is

$$\frac{f(\text{logit}^{-1}(u^*))|J(u^*)|g(u^{(t)} | u^*)}{f(\text{logit}^{-1}(u^{(t)}))|J(u^{(t)})|g(u^* | u^{(t)})}$$

where $|J|$ is the absolute value of the Jacobian.

The top plot uses $\varepsilon \sim \text{unif}(-1, 1)$, while the bottom plot uses $\varepsilon \sim \text{unif}(-0.01, 0.01)$.



Gibbs Sampling

Let $x = (x_1, \dots, x_p)^T$ and $x_{-i} = (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_p)^T$.

Denote the conditional density of $X_i \mid X_{-i}$ by $f(x_i \mid x_{-i})$. A general Gibbs sampling procedure is as follows.

- Select starting values $x^{(0)}$ and set $t = 0$
- Generate, in turn

$$X_1^{(t+1)} \mid \dots \sim f(x_1 \mid x_2^{(t)}, \dots, x_p^{(t)})$$

$$X_2^{(t+1)} \mid \dots \sim f(x_2 \mid x_1^{(t+1)}, x_3^{(t)}, \dots, x_p^{(t)})$$

$$\vdots$$

$$X_p^{(t+1)} \mid \dots \sim f(x_p \mid x_1^{(t+1)}, \dots, x_{p-1}^{(t+1)})$$

where $\mid \dots$ denotes conditioning on the most recent updates to all other elements of x . It is called one **cycle**.

- Increment t and go to step 2

Example: Fur Seal Pup Capture-Recapture Study

The goal is to estimate the number of pups in a fur seal colony. The individuals captured during each census are released with a marker indicating their capture. Population size can be estimated on the basis of the history of capture and recapture data.

Census i	1	2	3	4	5	6	7
Number captured c_i	30	22	29	26	31	32	35
Number newly caught m_i	30	8	17	7	9	8	5

We assume

- The population is closed, which means that deaths, births, and migrations are inconsequential during the sampling period.
- All animals are equally catchable on any one capture occasion, but capture probabilities may change over time.

Notations

- N : unknown population size
- I : total number of census attempts
- $c = (c_1, \dots, c_I)$: numbers of captures
- $\alpha = (\alpha_1, \dots, \alpha_I)$: unknown capture probabilities
- r : total number of distinct animals captured during the study

The likelihood is

$$L(N, \alpha \mid c, r) \propto \frac{N!}{(N-r)!} \prod_{i=1}^I \alpha_i^{c_i} (1 - \alpha_i)^{N-c_i}$$

Assume the following prior distributions with $\theta_1 = \theta_2 = 0.5$,

$$f(N) \propto 1, \quad \alpha_i \sim \text{Beta}(\theta_1, \theta_2), \quad i = 1, \dots, I$$

The posterior distribution is

$$f(N, \alpha \mid c, r) \propto \frac{N!}{(N-r)!} \prod_{i=1}^I \alpha_i^{c_i} (1 - \alpha_i)^{N-c_i} \prod_{i=1}^I \alpha_i^{\theta_1-1} (1 - \alpha_i)^{\theta_2-1}$$

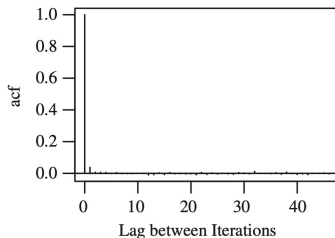
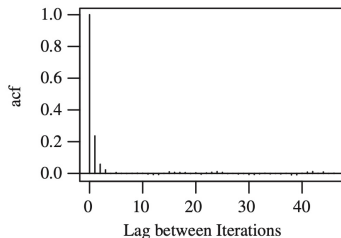
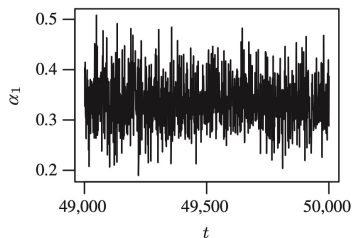
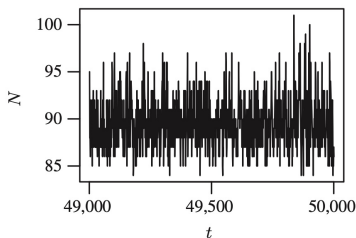
The conditional posterior distribution is

$$N \mid \alpha, \text{data} \sim \text{NegBin} \left(r, 1 - \prod_{i=1}^I (1 - \alpha_i) \right)$$

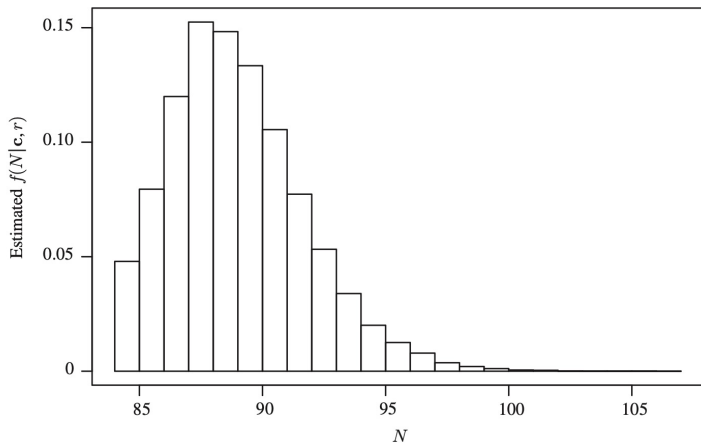
$$\alpha_i \mid N, \text{data} \sim \text{Beta}(c_i + \theta_1, N - c_i + \theta_2)$$

We draw 100,000 samples with the first 50,000 samples discarded for burn-in.

The trace plots and autocorrelation plots do not indicate any lack of convergence. The effective sample size is 45,206.



The posterior mean of N is 89.5 with a 95% HPD interval of (84, 94).



Properties of Gibbs Sampler

- Gibbs sampler is a special Metropolis-Hastings algorithm with acceptance ratio 1.
- The ordering of updates made to the components of X in the Gibbs sampler can change from one cycle to the next, which is called **random scan Gibbs sampling**. It can be effective when parameters are highly correlated.
- It is not necessary to treat each component of X individually. We may partition X into several blocks and update them sequentially.
- **Metropolis-within-Gibbs**: If one conditional distribution is not standard, we may apply Metropolis-Hastings algorithms for that step.

Fur Seal Pup Capture-Recapture Study

To mitigate the sensitivity due to the values of θ_1 and θ_2 , we further assume a prior for θ_1 and θ_2 ,

$$f(\theta_1, \theta_2) \propto \exp\{-(\theta_1 + \theta_2)/1000\}$$

Hence,

$$f(\theta_1, \theta_2 \mid \dots) \propto \left[\frac{\Gamma(\theta_1 + \theta_2)}{\Gamma(\theta_1)\Gamma(\theta_2)} \right]^I \prod_{i=1}^I \alpha_i^{\theta_1} (1 - \alpha_i)^{\theta_2} \exp \left\{ -\frac{\theta_1 + \theta_2}{1000} \right\}$$

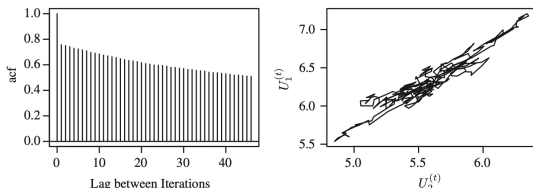
It is not a standard distribution, and we have to implement a Metropolis-Hastings algorithm to sample (θ_1, θ_2) .

Note that $\theta_1 > 0$ and $\theta_2 > 0$. Let $U = (U_1, U_2) = (\log \theta_1, \log \theta_2)$.

$$f(u_1, u_2 \mid \cdots) \propto f(\theta_1, \theta_2 \mid \cdots) \theta_1 \theta_2$$

Use a random walk with $\varepsilon \sim N(0, 0.085^2 I)$, where the acceptance rate is about 23%.

Draw 100,000 samples with the first 50,000 discarded. The Gelman-Rubin statistics for the parameters are close to 1, however, the autocorrelation plots indicate high correlation between iterates. The effective sample size is 1127.



Griddy-Gibbs Sampler

When it is difficult to sample from the univariate conditional density $f(x_k | x_{-k})$, we may select some grid points $z_1, \dots, z_n$ over the range of the support of $f(x_k | x_{-k})$. Let $w_j^{(t)} = f(z_j | x_{-k}^{(i)})$.

- Method 1: Approximate the CDF by linear interpolation and sample x_k by inverting the CDF
- Method 2: Draw x_k from the discrete distribution on $z_1, \dots, z_n$.

Comments.

- Updating the approximation from time to time makes the chain not time homogeneous.
- Treat the approximation as a proposal distribution, and accept the candidate value with probability.

Implementation

In theory, a Markov chain will converge to its stationary distribution under some regularity conditions. In practice, it is necessary to determine when the chain has run sufficiently long.

We will focus on the following issues:

- Graphical diagnostics
- Burn-in and run length
- Choice of proposal distributions
- Reparameterization
- Effective sample size
- Number of chains

Good Mixing and Convergence

We need to check a Markov chain for the following issues.

- How quickly the chain forgets its starting value
- Whether the chain has approximately reaches its stationary distribution
- How quickly the chain fully explores the support of the target distribution
- How far apart in a sequence observations need to be before they can be considered to be approximately independent

Graphical Diagnostics

- **Trace plot:** plot $X^{(t)}$ versus t . A chain that is mixing well will quickly move away from its starting value and the plot will wiggle about vigorously in the region supported by f .
- **Cumulative sum (cusum) diagnostics:** assess the convergence of an estimator of $\theta = E[h(X)]$. Let $\hat{\theta}_n = \sum_{i=1}^n h(X^{(i)})/n$. Plot $\sum_{i=1}^t [h(X^{(i)}) - \hat{\theta}_n]$ versus t . Cumsum plots that are very wiggly and have smaller excursions from 0 indicate that the chain is mixing well.
- **Autocorrelation plot.** plot the correlation in the sequence at different iteration lags. The autocorrelation at lag i is the correlation between iterates that are i iterations apart. A chain that has poor mixing will exhibit slow decay of the autocorrelation as the lag between iterations increases.

Gelman-Rubin Statistic

Gelman and Rubin proposed a testing procedure to assess the convergence of a Markov chain based on ANOVA.

- Suppose we have J runs of the MCMC algorithm to create separate, equal-length chains with starting values dispersed over the support of the target density.
- Let L be the length of chains after discarding D burn-in iterates.
- For the j th chain, the retained values are $x_j^{(D)}, \dots, x_j^{(D+L-1)}$.
- Let B and W be the between-chain variance and within-chain variance. Then compute

$$R = \frac{(L-1)W/L + B/L}{W}, \quad \hat{R} = \frac{J+1}{J}R - \frac{L-1}{JL}$$

In theory, $\sqrt{R} \rightarrow 1$ as $L \rightarrow \infty$. In practice, we use the adjusted R for better performance.

Comments

- $\sqrt{\hat{R}} < 1.1$ indicates that the burn-in and chain length are sufficient.
- Alternately, we can plot $\hat{R}$ versus the number of iterations. When $\hat{R}$ has not stabilized near 1, this suggests lack of convergence.
- If the chosen burn-in period did not yield an acceptable result, then increase D or increase L , or maybe both.
- A conservative choice is to use on-half of the iterations for burn-in.
- The performance of this diagnostic is improved if the iterates $x_j^{(t)}$ are transformed so that their distribution is approximately normal.

Choice of Proposal

- Choose a proposal distribution g that approximates the target distribution f well.
- If f/g is bounded, the convergence of the Markov chain to its stationary distribution is faster overall. Hence, choose a proposal distribution that is somewhat more diffuse than f .
- In practice, the variance of the proposal distribution can be selected through an informal iterative process.
 - Start a chain, and monitor the acceptance rate.
 - The optimal acceptance rate is in $[0.25, 0.50]$, with the best choice being about 0.44 for one-dimensional problems, and 0.234 for higher-dimensional problems.
 - To apply such rules, ensure that the target and proposal distributions are roughly normally distributed or at least simple, unimodal distributions.

Reparameterization

For a Gibbs sampler, performance is enhanced when components of X are as independent as possible.

For example, if f is bivariate normal with strong positive correlation. Suppose $Y = (X_1 + X_2, X_1 - X_2)$ yields two independent components. The chain will converge faster if working with Y .

If there are continuous covariates in a linear model, it is useful to center and scale the covariates to reduce correlation between the parameters in the model. For Bayesian linear models with random effects, hierarchical centering can be used to accelerate MCMC convergence.

It is typically adapted for specific models, and it is difficult to provide generic advice.

Hierarchical Centered Random Effects Model

In a study, pollutant levels are tested at different laboratories. Let y_{ij} be the pollutant level of the j th sample that was tested at the i th laboratory.

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, \dots, I, \quad j = 1, \dots, n_i.$$

Assume $\mu \sim N(\mu_0, \sigma_\mu^2)$, $\alpha_i \sim N(0, \sigma_\alpha^2)$, and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$.

We reparameterize the model as

$$y_{ij} = \gamma_i + \varepsilon_{ij}$$

where $\gamma_i = \mu + \alpha_i$ and $\gamma_i \mid \mu \sim N(\mu, \sigma_\alpha^2)$. Thus γ is centered about μ . Hierarchical centering usually produces better behaved MCMC chains when σ_ε^2 is not a lot larger than σ_α^2 .

Effective Sample Size

The effective sample size measures the severity of autocorrelation in a Markov chain. It can be estimated by

$$\frac{L}{1 + 2 \sum_{k=1}^{\infty} \rho(k)}$$

where L the number of iterations after burn-in and $\rho(k)$ is the autocorrelation between realizations that are k iterations apart.

- It is common to truncate the summation when $\hat{\rho}(k) < 0.1$.
- For a fixed number of iterations, an MCMC algorithm with a larger effective sample size is likely to converge more quickly.

Number of Chains

To diagnose whether or not a chain has become stuck in a one or more modes of the target distribution, we run multiple chains from diverse starting values and then compare the within- and between-chain behavior.

Multiple short-runs versus one long-run

- many short runs are more informative when they indicate poor convergence behavior
- a Markov chain will converge only after it is long enough

Do both if possible. Exploratory work can be carried out using several shorter runs. After building confidence that the implementation is a promising one, run one final very long run from a good starting point to calculate and publish results.

Reporting the Results

- If $\{X^{(t)}\}$ represents a p -dimensional Markov chain, then $X_j^{(t)}$ is a Markov chain whose limiting distribution is the j th marginal of f .
- Report standard summary statistics such as mean and variance. Discard the D burn-in iterates and estimate $E[h(X)]$ by

$$\frac{1}{L} \sum_{t=D}^{D+L-1} h(X^{(t)})$$

The estimator is consistent, though $X^{(t)}$ are serially correlated.

- Report quantile estimates and other interval estimates. Estimate them using the corresponding percentiles.
- Draw graphical summaries such as histograms, density curves, scatterplots, or other descriptive plots.

Monte Carlo Standard Error

The **Monte Carlo standard error** of an estimator measures the variability in the estimator if the MCMC algorithm were to be run repeatedly.

A simple estimator of the standard error is the **batch method**.

- Separate the L iterates into batches with b consecutive iterations in each batch.
- A recommended batch size is $b = \lfloor L^{1/a} \rfloor$, where $a = 2$ or 3 and $\lfloor z \rfloor$ denotes the largest integer less than z .
- Compute the mean of each batch.
- The estimated standard error is the standard deviation of these means divided by the square root of the number of batches.

It has been suggested that one should run the chain until the Monte Carlo simulation error is less than 5% of the standard deviation for all parameters of interest.

Highest Posterior Density

For a unimodal and symmetric posterior distribution, the $100(1 - \alpha)\%$ **highest posterior density** HPD interval is given by the $(\alpha/2)$ th and $(1 - \alpha/2)$ th percentiles of the iterates.

For a unimodal posterior distribution,

- Sort the MCMC realizations after burn-in, $x^{(D)}, \dots, x^{(D+L-1)}$ to obtain $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(L-1)}$.
- Compute the $100(1 - \alpha)\%$ credible intervals

$$I_j = (x_{(j)}, x_{(j + \lfloor (1-\alpha)(L-1) \rfloor)}),$$

for $j = 1, 2, \dots, (L-1) - \lfloor (1-\alpha)(L-1) \rfloor$

- The $100(1 - \alpha)\%$ HPD interval is the interval I_{j^*} with the shortest interval width among all credible intervals.

R Functions

The `coda` package provides some useful functions.

- `gelman.diag()`: Gelman and Rubin's convergence diagnostic
- `gelman.plot()`: plot Gelman-Rubin statistics against t
- `effectiveSize()`: compute effective sample size
- `batchSE()`: Monte Carlo standard error
- `HPDinterval()`: compute highest posterior density intervals